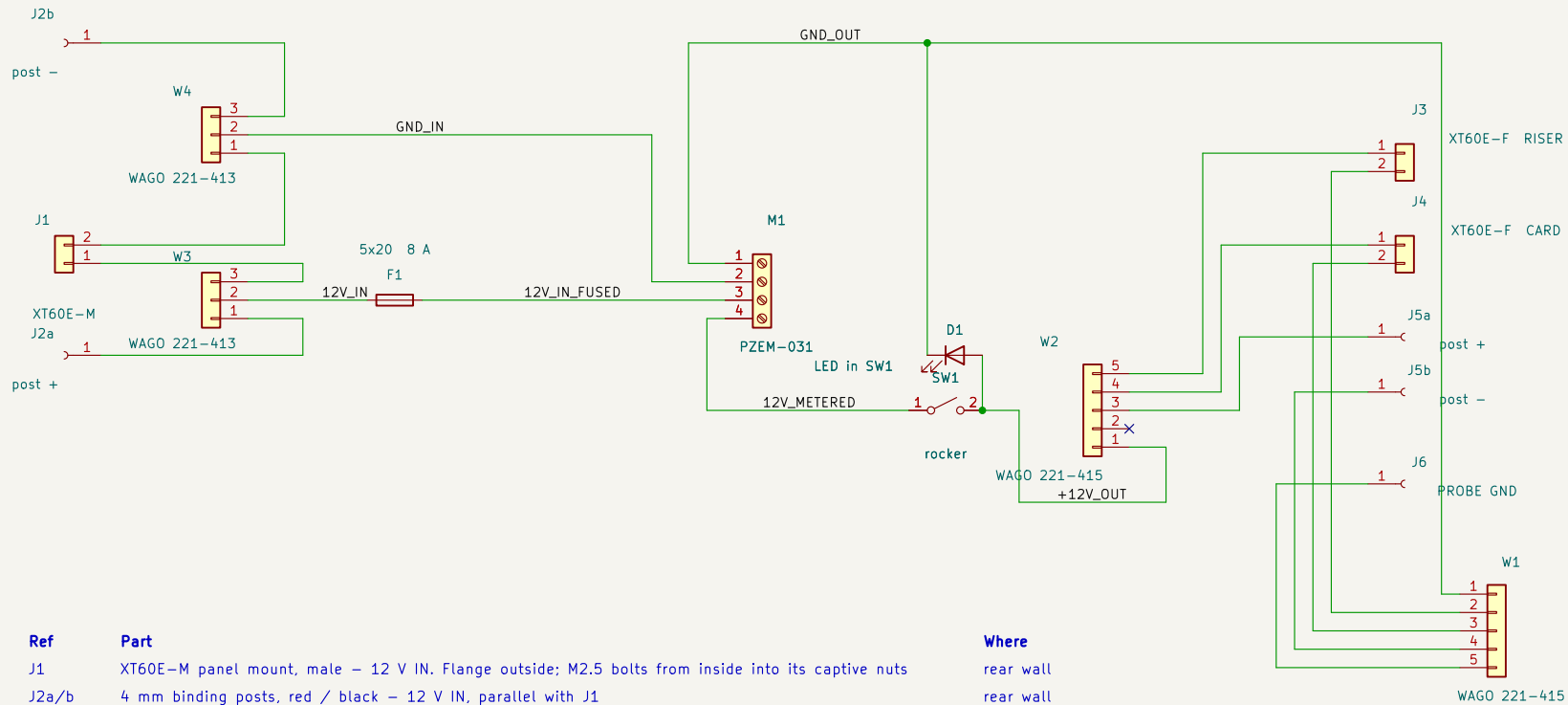


# PCIe GPU test rig – power box wiring

Rev D 2026-09-30. 12V IN -> F1 -> M1 -> SW1 -> outputs. Every join is a WAGO 221 lever nut: W3/W4 on the input pair, W2 on the output +, W1 the ground bus.



| Ref   | Part                                                                                              | Where      |
|-------|---------------------------------------------------------------------------------------------------|------------|
| J1    | XT60E-M panel mount, male – 12 V IN. Flange outside; M2.5 bolts from inside into its captive nuts | rear wall  |
| J2a/b | 4 mm binding posts, red / black – 12 V IN, parallel with J1                                       | rear wall  |
| W3    | WAGO 221-413 three-port: J1 +, J2a, one wire to F1                                                | inside     |
| W4    | WAGO 221-413 three-port: J1 –, J2b, one wire to M1 terminal 2                                     | inside     |
| F1    | 5x20 mm panel fuse holder, 8 A fast-blow glass fuse                                               | rear wall  |
| M1    | Peacefair PZEM-031 meter, 8-100 V / 0-20 A, built-in shunt                                        | top deck   |
| SW1   | Amppper 20 mm illuminated rocker, 12 V 20 A, 3-pin                                                | top deck   |
| D1    | the dot LED inside SW1 (yellow pin = LED –)                                                       | –          |
| W2    | WAGO 221-415 five-port: SW1 B in; RISER+, CARD+, OUT+ post out; one spare                         | inside     |
| W1    | WAGO 221-415 five-port, ground bus: M1 LOAD– in; RISER–, CARD–, OUT– post, PROBE out              | base floor |
| J3    | XT60E-F panel mount, female – OUT, RISER. Flange outside; M2.5 bolts + nuts                       | right wall |
| J4    | XT60E-F panel mount, female – OUT, CARD (optional 2nd, parallel)                                  | right wall |
| J5a/b | 4 mm binding posts, red / black – OUT +/-                                                         | right wall |
| J6    | 4 mm binding post, black – PROBE GND (scope / DMM reference)                                      | top deck   |

M1: the shunt is internal between 2 (DC IN-) and 1 (LOAD-). Input negative goes to 2 via W4, never to W1 – on the ground bus it bypasses the shunt and the meter reads 0 A.

SW1: A from M1.4, B to W2, yellow to ground (on the OUT – post terminal). Never swap B and yellow – dead short.

XT60E pigtails are 12 AWG, the top of the WAGO 221 range – push firmly. Runs 14 AWG. No solder in the box.

Supply: OWON at 12.0 V, OVP 12.6 V, 6 A current limit (the working max). F1 8 A fast 5x20. J4 is optional.

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